MANISH RAINA

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PROFILE

AI/ML Engineer with hands-on, end-to-end project experience across model development, backend engineering, and full-stack delivery. Proven track record shipping multi-agent orchestration systems, production ML pipelines, and full-stack apps with measurable accuracy, latency, and reliability gains using Python, Scikit-learn, TensorFlow, FastAPI, React, Docker, and RAG/MCP-based agent systems.

CORE COMPETENCIES

- **Languages:** Python, C, C++, JavaScript, SQL, TypeScript
- **AI Agents & Orchestration:** LangChain, LangGraph, Model Context Protocol (MCP), Multi-Agent Systems, Agentic Tool/Skill Design, Function Calling, Prompt Engineering, RAG Pipelines
- **ML & Model Training:** Scikit-learn, TensorFlow, LightGBM, NLP, Model Training & Fine-Tuning, Embedding-based Semantic Search, Model Evaluation (AUC, Cross-Validation, Calibration, IPCW Weighting)
- **Web & Backend / APIs:** FastAPI, React.js, REST APIs, API Design & Integration, WebSockets, JWT Authentication, RBAC, Middleware Validation
- **Data & Infra:** PostgreSQL, MongoDB, Redis, SQLite, Qdrant Vector DB, Docker, Docker-compose, Git
- **Other:** Web Scraping, Power BI, Speech Recognition & Synthesis, OpenCV, dlib, Playwright

EXPERIENCE

Machine Learning & Web Development Intern — EISystems Technologies
June 2024 – Aug 2024

- Contributed to data preprocessing and machine learning model development for live client projects, while building front-end features in React.js to support delivery.

Machine Learning Research Intern — NIT Delhi
July 2025 – Sept 2025

- Collaborated with a 3-person team to build a survival prediction model (LightGBM + IPCW) for Heart Disease outcomes, achieving 91.89% cross-validation AUC; conducted comparative performance analysis against a Cox Proportional Hazards baseline.

PROJECTS

Nexus AI OS v2.0 — Autonomous Multi-Agent Workspace & AI Orchestration Platform
GitHub

- Engineered a multi-agent orchestration kernel routing tasks across **12** specialized agents via a JSON-based DAG, supporting 12 task categories and 30+ sub-actions, with a custom MCP client integrating 6 MCP servers and 10 built-in tools.
- Achieved a **100%** pass rate on integration tests (4/4) and DAG workflows (6/6), with 96% task-step success (24/25); full 4-step workflows completed in 3.8–4.2 seconds.

AI Career Skill Intelligence Platform (Full-Stack + ML + Scraper)
GitHub

- Architected a full-stack skill-gap platform (FastAPI, React, PostgreSQL) with 39 REST endpoints, parsing 55 resumes at **98.2%** extraction accuracy and computing readiness scores in 65.3ms end-to-end.
- Processed 32,217 job listings across 3,947 companies, feature-engineering 94,766 skills and training a RandomForest skill-demand regressor (**R^2 = 0.9998**); generated readiness scores for 50+ candidates (avg 64.5%, top 92%).

AI Voice Receptionist (Speech-to-Speech AI Assistant)
GitHub

- Built a speech-to-speech voice assistant achieving **88%** intent recognition accuracy across 50 test queries, with 1.2–1.6s speech-to-response latency and offline pyttsx3 TTS generating responses in 80–120ms.
- Integrated a real-time face-recognition layer (DeepFace, VGG-Face, OpenCV) achieving **91.7%** accuracy across 60 live trials at ~450–650ms per inference frame, plus automated Gmail IMAP email briefing.

EDUCATION

B.Tech, Computer Science Engineering (AI & ML) — PIET, Haryana — *Aug 2022 – Aug 2026*

Senior Secondary (Class XII) — Kotwal National Institute — *Mar 2018 – Mar 2022*

CERTIFICATIONS & ACHIEVEMENTS

- Machine Learning with Python — Internship Certificate, EISystems Technologies
- Machine Learning Research Internship — NIT Delhi (2025)
- Advanced Diploma in Software Technology (12 Months) — Supertech (India) Computer Education